

Mathematical Analysis II Notes

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8 空间解析几何

Linear Algebra and Analytic Geometry

8.0.1 Inner Product

Let V be a real vector space, an inner product on V is a map.

$$\begin{aligned} g : V \times V &\rightarrow \mathbb{R} \\ (u, w) &\mapsto g(u, w) \end{aligned}$$

satisfying

1. **Bilinearity:**

$$\begin{aligned} g(\lambda_1 v_1 + \lambda_2 v_2, w) &= \lambda_1 g(v_1, w) + \lambda_2 g(v_2, w) \\ g(v, \lambda_1 w_1 + \lambda_2 w_2) &= \lambda_1 g(v, w_1) + \lambda_2 g(v, w_2) \end{aligned}$$

2. **Symmetric:** $g(v, w) = g(w, v), \forall (v, w) \in V$

3. **Positive Definite 正定:** $g(v, v) \geq 0, \forall v, “=”$ iff $v = 0$

An inner product is also denoted by $\langle \cdot, \cdot \rangle_g$, or *angle* $\langle \cdot, \cdot \rangle$, or “ $\cdot$ ” if there is no confusion.

Eg. $V = C[0, 1], \langle f, g \rangle \triangleq \int_0^1 f(x)g(x) dx$

For $\dim_{\mathbb{R}} V = n < +\infty$, we have the following classification results:

Thm (Classification of **Symmetric Bilinear Form**). *If $\dim_{\mathbb{R}} V = n$, then for any symmetric bilinear form g on V , there exists a basis $(v_1, \dots, v_n)$ s.t.*

$$(g(v_i, v_j))_{1 \leq i, j \leq n} = \begin{bmatrix} 1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 1 & & & & & & \\ & & & -1 & & & & & \\ & & & & \ddots & & & & \\ & & & & & -1 & & & \\ & & & & & & 0 & & \\ & & & & & & & \ddots & \\ & & & & & & & & 0 \end{bmatrix} (r_+ \uparrow 1, r_- \uparrow -1, n - r_+ - r_- \uparrow 0)$$

where $r_+ = \max \{ \dim_{\mathbb{R}} W \mid g|_W > 0 \text{ for a subsequence } W \}$, r_- similarly defined.

is called the positive/negative inertia index respectively.

Thm (Classification of Inner Product). *If $\dim_{\mathbb{R}} V = n$, then for any inner product g on V ,*

1. there exists a basis $(v_1, \dots, v_n)$ s.t.

$$g(v_i, v_j) = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}, \text{ i.e. } g(v_i, v_j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \text{ (**Kronecker Symbol**)}$$

i.e. orthonormal basis w.r.t. g exists.

2. any basis $(v'_1, \dots, v'_n)$ is orthonormal w.r.t. g iff. $(v'_1, \dots, v'_n) = (v_1, \dots, v_n)P$ for some $P \in O(n) = \{A \in M_{n \times n}(\mathbb{R}) \mid AA^T = I\}$, **called the orthonormal group 正交群**

In coordinates

Let V be a n -dim real vector space, an ordered set of n linearly independent vectors is called an ordered basis, usually denoted by $(v_1, \dots, v_n)$.

Fix a particular choice of ordered basis $(v_1, \dots, v_n)$, we have one-to-one correspondence.

1. $V \rightarrow \mathbb{R}^n$ **isomorphism 同构**

$$v \rightarrow x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ coordinate vector (w.r.t. the ordered basis } v_1, \dots, v_n)$$

$$\text{s.t. } v = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

2.

$L(v) = \{\text{linear map from } V \text{ to } V\} \rightarrow M_{n \times n}(\mathbb{R})$ **isomorphism**

$\mathcal{A} \mapsto A$ representing the linear map V under the basis $(v_1, \dots, v_n)$

$$\mathcal{A} \cdot (\mathcal{B} + \mathcal{C}) = \mathcal{A} \cdot \mathcal{B} + \mathcal{A} \cdot \mathcal{C} \quad A(B + C) = AB + AC$$

3.

$\{\text{Bilinear form on } V\} \rightarrow M_{n \times n}(\mathbb{R})$ **isomorphism**

$$g \mapsto (g(v_i, v_j))_{1 \leq i, j \leq n}$$

4.

$\{\text{Symmetric Bilinear form on } V\} \rightarrow \text{Sym}_{n \times n}(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) \mid A^T = A\}$ **isomorphism**

$$g \mapsto (g(v_i, v_j))_{1 \leq i, j \leq n}$$

5. $\{\text{inner product on } V\} \rightarrow \text{Sym}_{n \times n}^+(\mathbb{R}) = \{A \in \text{Sym}_{n \times n}(\mathbb{R}) \mid A > 0\}$

(5) $\subseteq$ (4) **convex cone** $\rightarrow g_1, g_2 \Rightarrow \lambda_1 g_1 + \lambda_2 g_2, \lambda_1 \lambda_2 \geq 0$, but $\lambda_1^2 + \lambda_2^2 \neq 0, \dots$

Thm. For any $A \in \text{Sym}_{n \times n}(\mathbb{R})$, there exists $P \in GL(n, \mathbb{R}) = \{B \in M_{n \times n}(\mathbb{R}) \mid \det B \neq 0\} \rightarrow$
General Linear Group $GL(n)$

$$s.t. PAP^T = \begin{pmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 1 & & & & & \\ & & & -1 & & & & \\ & & & & \ddots & & & \\ & & & & & -1 & & \\ & & & & & & 0 & \\ & & & & & & & \ddots \\ & & & & & & & & 0 \end{pmatrix}$$

Thm. For any $A \in \text{Sym}_{n \times n}^+(\mathbb{R})$, there exists $P \in GL(n, \mathbb{R})$, s.t. $PAP^T = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$

$$(\Rightarrow A = P^{-1} (P^{-1})^T)$$

Thm (Spectral Theorem / Orthogonal Diagonalization). For any $A \in \text{Sym}_{n \times n}(\mathbb{R})$, there exists $P \in O(n)$ and $\lambda_1, \dots, \lambda_n \in \mathbb{R}$, s.t. $PAP^T = \text{diag}(\lambda_1, \dots, \lambda_n) = A$ ($\Rightarrow$ **If $A > 0$, $A = P^T \Lambda P = P^T \Lambda^{\frac{1}{2}} P \cdot P^T \Lambda^{\frac{1}{2}} P = B^2$, where $B^T = B$, $B > 0$**)

8.0.2 Geometric notions induced from inner product g on V

1. Length of a vector: $v \in V, |v|^g = \sqrt{\langle v, v \rangle_g}$
2. Angle: for $v, w \in V \setminus \{0\}$, the angle between v and w is defined as

$$\theta(v, w) = \cos^{-1} \frac{\langle v, w \rangle_g}{|v|_g |w|_g}$$

(where $\cos^{-1}$ is the inverse of $\cos : [0, \pi] \rightarrow [-1, 1]$)

(Cauchy-Schwarz Inequality: $|\langle v, w \rangle_g| \leq |v|_g |w|_g$)

If $\theta(v, w) = \frac{\pi}{2}$, we say v is perpendicular to w w.r.t. g , denoted by $v \perp_g w$, or $v \perp w$ if no confusion.

3. Orthogonal projection: for $\vec{b} \in V \setminus \{0\}$



$$\text{pr}_{\vec{b}} : V \rightarrow \langle \vec{b} \rangle$$

$$\vec{a} \mapsto \frac{\langle \vec{a}, \vec{b} \rangle_g}{|\vec{b}|_g} \frac{\vec{b}}{|\vec{b}|_g}$$

$$\text{pr}_{\langle \vec{b} \rangle^\perp} : V \rightarrow \langle \vec{b} \rangle^{\perp_g} \text{ orthogonal complement}$$

$$\vec{a} \mapsto \vec{a} - \text{pr}_{\vec{b}} \vec{a}$$

$$V = \langle \vec{b} \rangle \oplus \langle \vec{b} \rangle^{\perp_g}$$

4. Volume of “ k -dimensional” parallelepiped 平行多面体 is defined inductively.

The set $P_{v_1, \dots, v_k} = \left\{ \sum_{i=1}^k t_i v_i \mid 0 \leq t_1, \dots, t_n \leq 1 \right\}$ is called a parallelepiped spanned by $(v_1, \dots, v_k)$ in V .

- $k = 1$, $\text{Vol}_g(P_v) = |v|_g$
- $k = 2$, $\text{Vol}_g(P_{v_1, v_2}) \triangleq \text{Vol}_g(P_{v_1}) \left| \text{pr}_{\langle v_1 \rangle^\perp} v_2 \right|_g$

$$\begin{aligned} \text{Vol}_g(P_{v_1, v_2})^2 &= |v_1|_g^2 \left(|v_2|_g^2 - \left(\frac{\langle v_1, v_2 \rangle_g}{|v_2|_g} \right)^2 \right) \\ &= |v_1|_g^2 |v_2|_g^2 - \langle v_1, v_2 \rangle_g^2 \\ &= \det \begin{pmatrix} \langle v_1, v_1 \rangle_g & \langle v_1, v_2 \rangle_g \\ \langle v_2, v_1 \rangle_g & \langle v_2, v_2 \rangle_g \end{pmatrix} \end{aligned}$$

$$v_2 = \text{pr}_{v_1} v_2 + \text{pr}_{v_1^\perp} v_2$$

$$|v_2|_g^2 = |\text{pr}_{v_1} v_2|_g^2 + |\text{pr}_{v_1^\perp} v_2|_g^2$$

$$\text{Thm. } \text{Vol}_g(P_{v_1, \dots, v_k}) = \sqrt{\det \left(\langle v_i, v_j \rangle_g \right)}$$

8.0.3 Orientation 定向

Def (Equivalence Relation). A subset $S \subseteq X \times X = \{(x, y) \mid x \in X, y \in Y\}$ is called a binary relation on X . A binary relation S on X is called an **equivalence relation** 等价关系 on X if

1. Symmetry: $(x, y) \in S \Rightarrow (y, x) \in S$
2. Transitivity: $(x, y) \in S, (y, z) \in S \Rightarrow (x, z) \in S$

3. *Reflexivity*: $(x, x) \in S, \forall x \in X$

It is usually denoted by “ $\sim_S$ ” or “ $\sim$ ”

Def (Equivalence Class). Let “ $\sim$ ” be equivalence relation on X , then for any $x \in X$, the set $\{y \in X \mid y \sim x\} \subseteq X$ is called an equivalence class, it is denoted by $[x]_\sim$. Any $y \in [x]_\sim$ is called a representative of $[x]_\sim$. $X/\sim \triangleq \{[x]_\sim \mid x \in X\}$ is called the quotient set of X under the relation “ $\sim$ ”.

Eg. 1. $\{(x, y) \in \mathbb{R}^2 \mid x > y\}$ (“ $>$ ”) is a binary relation on $\mathbb{R}$, but not an equivalence relation.

2. $S = \Delta x = \{(x, x) \mid x \in X\}$ (“ $=$ ”) is an equivalence relation.

3. (Congruence) Fix $n \in \mathbb{N}$

$$\{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid n \mid (x - y)\}$$

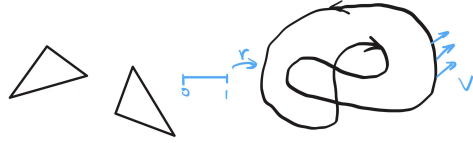
This equivalence relation is called “ $\text{mod } n$ congruence” ($\equiv n$)

$$\mathbb{Z}/\equiv n \left(\triangleq \mathbb{Z}_n \right) = \{[0]_{\equiv n}, \dots, [n-1]_{\equiv n}\}$$

$$\mathbb{Z}_2 = \{[0]_{\equiv 2}, [1]_{\equiv 2}\}$$

$[k]_{\equiv n} = \{l \in \mathbb{Z} \mid n \mid (l - k)\} = \{k + mn \mid m \in \mathbb{Z}\}$, is called the “ $\text{mod } n$ congruence class of k ”.

Eg. 1. $X = \{\text{triangle on } \mathbb{R}^2\} \cong \cong$



2. “Similarity” of matrix $A \sim B \iff A = PBP^{-1}$ for some $P \in GL(n, \mathbb{R})$

$$M_{n \times n}(\mathbb{R})/\sim = \{\text{Jordan canonical form}\}$$

Def (Orientation 定向, Oriented Vector Space 定向线性空间). Let $\mathcal{B}_V = \{\text{Ordered basis of } V\}$ where V is a real vector space, define

$$(v'_1, \dots, v'_n) = (v_1, \dots, v_n)P$$

for some $P \in GL(n, \mathbb{R})$, $\det P > 0$ iff $(v'_1, \dots, v'_n) \sim (v_1, \dots, v_n)$.

This is an equivalence relation, and $B_V/\sim = \{o_1, o_2\}$, an element of $B_V/\sim$ is called an orientation on V . Any vector space, together with a choice of orientation, is called an oriented vector space.

8.0.4 $\mathbb{R}^n$: Euclidean Space

Equipped with the standard inner product

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

where $x = (x_1, \dots, x_n)^+, y = (y_1, \dots, y_n)^+ \in \mathbb{R}^n$.

Under this inner product, $(e_1, \dots, e_n)$, where $e_i = (0, \dots, 0, \overset{i}{1}, 0, \dots, 0)$, is called the *standard orthonormal basis*.

Rmk. $n = 2$, (e_1, e_2) is usually denoted by i, j ;

$n = 3$, (e_1, e_2, e_3) is denoted by i, j, k

Def (Cross Product in $\mathbb{R}^3$). Let $x, y \in \mathbb{R}^3$, define " $\times$ " via

$$\begin{aligned} x \times y &= \begin{vmatrix} i & j & k \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix} i - \begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix} j + \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} k \\ &= (x_2 y_3 - x_3 y_2) i + (x_3 y_1 - x_1 y_3) j + (x_1 y_2 - x_2 y_1) k \end{aligned}$$

call it the cross product of x and y .

Prop. 1. $x \times y = (-y \times x)$ ($x \times y = 0 \iff \{x, y\}$ linear dependent) (*Anti-Symmetric*)

2. $(\lambda x + \mu y) \times z = \lambda x \times z + \mu y \times z$ (*Bilinear*)

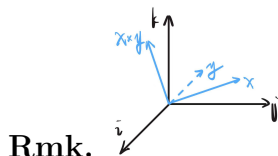
3. $\langle x \times y, x \rangle = 0$ $\langle x \times y, y \rangle = 0$; $(x, y, x \times y)$ determines the same orientation as (i, j, k)

4. $|x \times y| = \text{Area}(P_{x,y})$

Pf. 3.

$$\begin{aligned} (x, y, x \times y) &= (i, j, k) \begin{vmatrix} x_1 & y_1 & x_2 & y_2 \\ x_2 & y_2 & x_3 & y_3 \\ x_3 & y_3 & x_1 & y_1 \\ x_1 & y_1 & x_2 & y_2 \end{vmatrix} \\ &= |x \times y|^2 \end{aligned}$$

$\implies$ if $\{x, y\}$ are linearly independent, $(x, y, x \times y)$ has the same orientation as (i, j, k) .



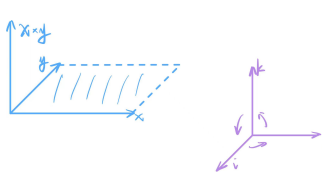
Rmk.

- The orientation denoted by (i, j, k) is called *Right-Hand orientation*;
- The orientation denoted by $(i, j, -k)$ is called *Left-Hand orientation*;
- The ordered basis with the *Right-Hand orientation* is called a *Right-Hand system*.

4.

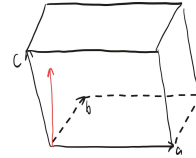
$$\begin{aligned}
 |x \times y|^2 &= (x_2y_3 - x_3y_2)^2 + (x_3y_1 - x_1y_3)^2 + (x_1y_2 - x_2y_1)^2 \\
 \text{Area}(P_{x,y}) &= \det \begin{pmatrix} \langle x, x \rangle & \langle x, y \rangle \\ \langle y, x \rangle & \langle y, y \rangle \end{pmatrix} \\
 &= (x_1^2 + x_2^2 + x_3^2)(y_1^2 + y_2^2 + y_3^2) - (x_1y_1 + x_2y_2 + x_3y_3)^2
 \end{aligned}$$

Thm. In $\mathbb{R}^3$, the cross product is uniquely determined by the four properties.



$$\begin{cases} i \times j = k \\ j \times k = i \\ k \times i = j \end{cases} \implies x \times y = \dots$$

Def (Mixed Product on $\mathbb{R}^3$ 混合积). Let $a, b, c \in \mathbb{R}^3$

$$\langle a, b \times c \rangle = \langle b, a \times c \rangle = \langle c, a \times b \rangle = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$


is called the mixed product of a, b, c .

- $\langle c, a \times b \rangle$ measures the “**signed volume**” of $P_{a,b,c}$;
- This equation also gives the formula for the volume of a parallelepiped in $\mathbb{R}^3$;
- $\langle c, a \times b \rangle \begin{cases} > 0 & (a, b, c) \text{ is a Right-Hand system} \\ = 0 & \{a, b, c\} \text{ linearly dependent} \\ < 0 & (a, b, c) \text{ is a Left-Hand system} \end{cases}$

Recall:

$$\text{Vol}(P_{v_1, \dots, v_k}) = \sqrt{\det(\langle v_i, v_j \rangle)}, \quad v_1, \dots, v_k \in \mathbb{R}^n$$

If $v_i = \sum_{s=1}^n e_s P_{s_i}$ ($\iff (v_1, \dots, v_k) = (e_1, \dots, e_n) P$), then

$$\begin{aligned}
 \langle v_i, v_j \rangle &= \left\langle \sum_{s=1}^n e_s P_{s_i}, \sum_{t=1}^n e_t P_{t_j} \right\rangle = \sum_{s,t=1}^n P_{s_i} P_{t_j} \delta_{st} = \sum_{s=1}^n P_{s_i} P_{s_j} \\
 &= (P^T P)_{ij}
 \end{aligned}$$

$$\implies \text{Vol}(P_{v_1, \dots, v_k}) = \sqrt{\det(P^T P)}$$

$$(\implies \text{In } \mathbb{R}_3, \text{Vol}(P_{a,b,c}) = \sqrt{\det(P^T P)} = \sqrt{(\det P)^2} = |\det P|)$$

Thm (Cauchy-Binet).

$$\det(U^T U) = \sum_{\substack{1 \leq i_1 < \dots < i_k \leq n \\ \binom{n}{k}}} U \begin{pmatrix} i_1 & \dots & i_k \\ 1 & \dots & k \end{pmatrix}^2$$

$$U^T U = \begin{pmatrix} U_{11} & \dots & U_{n1} \\ \vdots & & \vdots \\ U_{1k} & \dots & U_{nk} \\ \uparrow \\ i_1 & i_k \end{pmatrix} \begin{pmatrix} U_{11} & \dots & U_{1k} \\ \vdots & & \vdots & \leftarrow i_1 \ i_k \\ U_{n1} & \dots & U_{nk} \end{pmatrix}$$

Corollary (Generalized Pythagorean Theorem).

$$\text{Vol}(P_{v_1, \dots, v_k})^2 = \sum_{1 \leq i_1 < \dots < i_k \leq n} U \begin{pmatrix} i_1 & \dots & i_k \\ 1 & \dots & k \end{pmatrix}^2 \quad (k = 1, |V|^2 = \langle V, e_1 \rangle^2 + \dots + \langle V, e_n \rangle^2)$$

8.1 向量与坐标系

8.1.1 向量的定义与向量的加法和数乘

8.1.2 向量的共线和共面

8.1.3 向量的点乘和叉乘

8.1.4 向量的坐标表示

8.1.5 空间坐标系

8.2 Lines and Planes in $\mathbb{R}^3$

8.2.1 Plane

2-dim'l affine subspace.

- (General Equation) $\Pi : ax + by + cz + d = 0$ **Implicit equation**

Suppose $r_0 = (x_0, y_0, z_0) \in \Pi$, i.e. $ax_0 + by_0 + cz_0 + d = 0$

$$\begin{aligned} ax + by + cz + d = 0 &\iff a(x - x_0) + b(y - y_0) + c(z - z_0) = 0 \\ &\iff \langle \vec{r} - \vec{r}_0, \vec{n} \rangle = 0 \text{ where } \vec{n} = (a, b, c)^T \\ &\quad \text{(Point-Norm Form 点法式)} \end{aligned}$$

- (Point-Directions Form) Take $\vec{r}_0 \in \Pi$, $Q, R \in \Pi$, s.t. $\overrightarrow{PQ}, \overrightarrow{PR}$ linearly independent,
 $\vec{u} = \overrightarrow{PQ}$, $\vec{v} = \overrightarrow{PR}$, then Π can be explicitly parameterized by

$$\vec{r} = \vec{r}_0 + s\vec{u} + t\vec{v}$$

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} \text{ normal vector to } \Pi$$

8.2.2 Line

1-dim'l affine subspace, i.e. shift of 1-dim'l vector subspace.

- (Point-Direction Form) $\vec{r} = \vec{r}_0 + t\vec{v}$, $t \in \mathbb{R}$ **Explicit equation 显式表达**
- (Plane-Intersection) The set of $\mathbb{R}^3$ defined by $\begin{cases} a_1x + b_1y + c_1z + d_1 = 0 \\ a_2x + b_2y + c_2z + d_2 = 0 \end{cases}$, satisfying
 $\text{rk} \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix} = 2$. **Implicit equation 隐式表达**

- (1) $\implies$ (2) $(\vec{r} - \vec{r}_0) \times \vec{v} = \vec{0}$

$$\begin{aligned} \begin{pmatrix} i & j & k \\ x - x_0 & y - y_0 & z - z_0 \\ l & m & n \end{pmatrix} = 0 &\implies \begin{cases} n(y - y_0) = m(z - z_0) \\ l(z - z_0) = n(x - x_0) \\ m(x - x_0) = l(y - y_0) \end{cases} \\ &\implies \left(\begin{array}{ccc|c} 0 & n & -m & * \\ -n & 0 & l & * \\ m & l & 0 & * \end{array} \right) \end{aligned}$$

- (2) $\implies$ (1) $\xrightarrow{\text{Particular solution}} \vec{r}_0 + t(\vec{n}_1 \times \vec{n}_2)$ where $\vec{n}_1 = (a_1, b_1, c_1)^T$, $\vec{n}_2 = (a_2, b_2, c_2)^T$

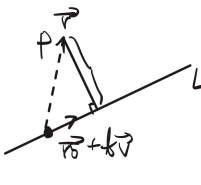
8.2.3 Relation between Lines & Planes

1. Two planes:
$$\begin{cases} \Pi_1 : a_1x + b_1y + c_1z + d_1 = 0 & \rightsquigarrow \vec{n}_1 \\ \Pi_2 : a_2x + b_2y + c_2z & \rightsquigarrow \vec{n}_2 \end{cases}$$

(a) Parallel: $\vec{n}_1 \parallel \vec{n}_2$

(b) Intersecting: " $\theta = \arccos \frac{\langle \vec{n}_1, \vec{n}_2 \rangle}{|\vec{n}_1| |\vec{n}_2|}$ "

2. Point & line:
$$d(P, L) = \frac{|(\vec{r} - \vec{r}_0) \times \vec{v}|}{|\vec{v}|}$$



Point & plane: $\Pi : ax + by + cz + d = 0 \implies d(P, \Pi) = \frac{|ax + by + cz + d|}{\sqrt{a^2 + b^2 + c^2}}$

$$P = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

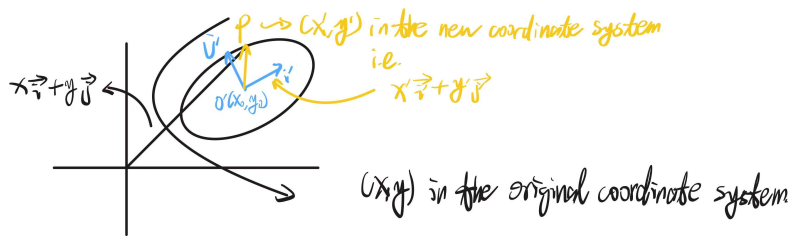
$$d(P, \Pi) = (\vec{r} - \vec{r}_0) \cdot \vec{n} = \frac{(a(x-x_0) + b(y-y_0) + c(z-z_0))}{\sqrt{a^2 + b^2 + c^2}}$$

3. Two lines:
$$\begin{cases} L_1 : \vec{r} = \vec{r}_1 + t\vec{v}_1 \\ L_2 : \vec{r} = \vec{r}_2 + t\vec{v}_2 \end{cases}$$

(a) Co-planar $\iff \langle \vec{r}_1 - \vec{r}_2, \vec{v}_1 \times \vec{v}_2 \rangle = 0$

(b) Distance: $\left\langle \vec{r}_2 - \vec{r}_1, \frac{\vec{v}_1 \times \vec{v}_2}{|\vec{v}_1 \times \vec{v}_2|} \right\rangle$

8.3 Conic Section/Curve & Quadric Surface



$$\Rightarrow \begin{pmatrix} \vec{i} & \vec{j} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \overrightarrow{OP} = \overrightarrow{OO'} + \overrightarrow{O'P}$$

$$= \begin{pmatrix} \vec{i} & \vec{j} \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} \vec{i}' & \vec{j}' \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$= \begin{pmatrix} \vec{i} & \vec{j} \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} + \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = P \begin{pmatrix} x' \\ y' \end{pmatrix} + \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \text{ coordinate transformation}$$

Quadric in $\mathbb{R}^n$

$$f(x_1, \dots, x_n) = \sum_{i,j=1}^n x_i A_{ij} x_j + \sum_{i=1}^n b_i x_i + C, \text{ where } A = A^T$$

$$= X^T A X + b^T X + C, \text{ where } X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Question: Can we find coordinate transformation $x = Py$ where $P \in O(n)$ to simplify the expression:

$$y^T P^T A P y + b^T P y + C$$

1. Take P s.t.

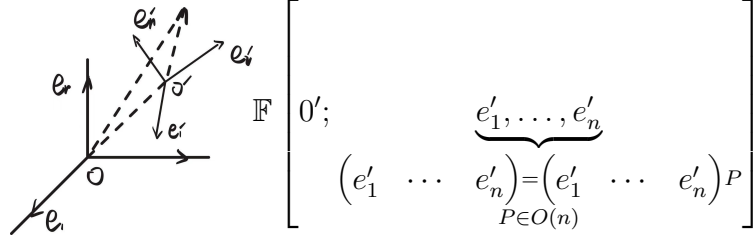
$$P^T A P = \begin{pmatrix} \lambda_1 & & & & & & \\ & \ddots & & & & & \\ & & \lambda_{r_+} & & & & \\ & & & -\lambda_{r_++1} & & & \\ & & & & \ddots & & \\ & & & & & \lambda_{r_++r_-} & \\ & & & & & & 0 \\ & & & & & & & \ddots \\ & & & & & & & & 0 \end{pmatrix}$$

2. $y = z + \begin{pmatrix} \vdots \\ \vdots \end{pmatrix}$

$$3. \lambda_1 z_1^2 + \cdots + \lambda_{r_+} z_{r_+}^2 - \left(\lambda_{r_++1} z_{r_++1}^2 + \cdots \right)$$

$$\text{Quadric surface: } \left\{ \vec{x} \in \mathbb{R}^n \mid f(\vec{x}) = \vec{x}^T A \vec{x} + \vec{b}^T \vec{x} + c = 1 \right\}, A = A^T$$

Descartes / Catesian Coordinate



A point has coordinate

$$\left\{ \begin{array}{l} \vec{x}' = \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix} \text{ in } \mathbb{F}[0'; e'_1, \dots, e'_n] \\ \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ in } \mathbb{F}[0; e_1, \dots, e_n] \end{array} \right. \Rightarrow \begin{array}{l} \vec{OO'} \\ = (e_1 \dots e_n) \vec{v} \end{array} + \begin{pmatrix} e'_1 & \dots & e'_n \end{pmatrix} \vec{x}' = \begin{pmatrix} e_1 & \dots & e_n \end{pmatrix} \vec{x}$$

$$\Rightarrow \vec{x} = \vec{v} + p \vec{x}'$$

Step 1. $\vec{x} = p \vec{x}' \Rightarrow x' P^T A P x' + b^T P x' + c = 0$
 对角阵

Find P s.t.

$$\xrightarrow{\text{orthogonal diagonalization}} P^T A P = \Lambda = \text{diag} \left(\lambda_1, \dots, \lambda_{r_+}, -\lambda_{r_++1}, \dots, -\lambda_{r_++r_-}, \underbrace{0, \dots, 0}_{n-\text{rank } A} \right)$$

$$\text{then } \lambda_1 x_1'^2 + \dots + \lambda_{r_+} x_{r_+}'^2 - \lambda_{r_++1} x_{r_++1}'^2 - \dots - \lambda_{r_++r_-} x_{r_++r_-}'^2 + \mathbf{b}'_1 \mathbf{x}'_1 + \dots + \mathbf{b}'_n \mathbf{x}'_n + c = 0$$

Step 2. Completing square for the variables $x'_1, \dots, x'_{r_+}, x'_{r_++1}, \dots, x'_{r_++r_-}$

$$\begin{pmatrix} x'_1 \\ \vdots \\ x'_{r_+} \\ x'_{r_++1} \\ \vdots \\ x'_{r_++r_-} \\ \vdots \\ x'_n \end{pmatrix} = \begin{pmatrix} x''_1 - \frac{b'_1}{2\lambda_1} \\ \vdots \\ x''_{r_+} - \frac{b'_{r_+}}{2\lambda_{r_+}} \\ x''_{r_++1} - \frac{b'_{r_++1}}{2\lambda_{r_++1}} \\ \vdots \\ x''_{r_++r_-} - \frac{b'_{r_++r_-}}{2\lambda_{r_++r_-}} \\ x''_{r_++r_-+1} \\ \vdots \\ x''_n \end{pmatrix}$$

i.e. $x'_1 = x'' + b''$

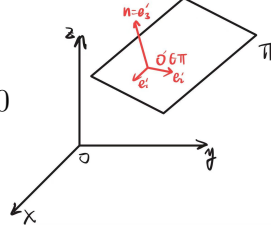
$$\lambda_1 x_1''^2 + \dots + \lambda_{r_+} x_{r_+}''^2 - \lambda_{r_++1} x_{r_++1}''^2 - \dots - \lambda_{r_++r_-} x_{r_++r_-}''^2 + b'_{r_++r_-+1} x_{r_++r_-+1}'' + \dots + b'_n x_n'' + c = 0$$

Step 3. Case 1.

$$b'_{r+1} = \dots = b'_n = 0 \rightsquigarrow x''^T \text{diag}(\lambda_1, \dots, \lambda_{r_+}, -\lambda_{r_++1}, \dots, -\lambda_{r_++r_-}, 0, \dots, 0) x'' + c = 0$$

Case 2. We can do another orthogonal transformation on the variable $x''_{r+1}, \dots, x''_n$ to simplify the 1st-order term s.t. the 1st order term becomes $b'''_{r+1} x'''_{r+1}$

Back to plane in $\mathbb{R}^3$: $ax + by + cz + d = 0$



Let $e'_3 = n = \frac{(a,b,c)}{\sqrt{a^2+b^2+c^2}}$, e'_1, e'_2 parallel to Π s.t. (e'_1, e'_2, n) is an orthonormal basis, then in the Cartesian coordinate $\mathbb{F}[0'; e'_1, e'_2, e'_3]$, Π is defined by $z' = 0$

$$\begin{pmatrix} a & b & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0$$

Let $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = P \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} + q$ (Euclidean transformation), then the equation is changed to

$$\begin{pmatrix} a & b & c \end{pmatrix} P \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} + \begin{pmatrix} a & b & c \end{pmatrix} q + d = 0$$

find $P \in O(3)$, $q \in \mathbb{R}^3$ s.t. $\begin{pmatrix} a & b & c \end{pmatrix} P = \begin{pmatrix} 0 & 0 & c \end{pmatrix}$, $c \neq 0$, $\begin{pmatrix} a & b & c \end{pmatrix} q + d = 0$

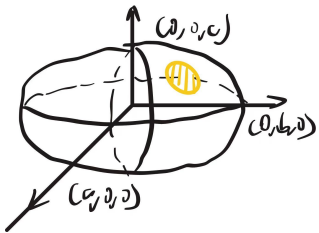
Thm. Given $S = \{ \vec{x} \in \mathbb{R}^n \mid f(\vec{x}) = \vec{x}^T A \vec{x} + \vec{b}^T \vec{x} + c = 0 \}$, we can find a coordinate transformation of the form $\vec{x} = P \vec{y} + \vec{d}$, where $P \in SO(n) = \{ Q \in O(n) \mid \det Q = 1 \}$ and $\vec{d} \in \mathbb{R}^n$, such that the equation $f(\vec{x}) = 0$ becomes

$$(1) \sum_{i=1}^{r_+} \lambda_i y_i^2 - \sum_{j=r_++1}^{r_++r_-} \lambda_j y_j^2 = c \text{ or}$$

$$(2) \sum_{i=1}^{r_+} \lambda_i y_i^2 - \sum_{j=r_++1}^{r_++r_-} \lambda_j y_j^2 - y_{r_++r_-+1} = 0$$

Classification of Quadric Surface in $\mathbb{R}^3$

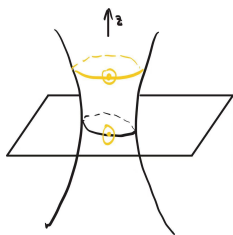
1. rank $A = 3$

$$(+, +, +; +) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$


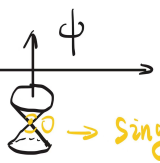
ellipsoid

$$(+, +, +; 0) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0$$

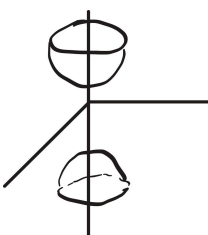
$$(+, +, +; -) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = -1$$

$$(+, +, -; +) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$


single sheeted hyperboloid 单叶双曲面

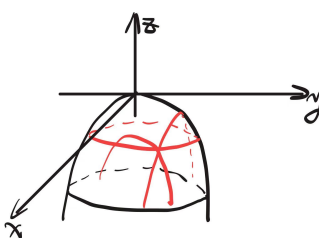
$$(+, +, -; 0) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$


$\rightarrow$ Singularity 奇点

$$(+, +, -; -) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$


Double sheeted hyperboloid 双叶双曲面

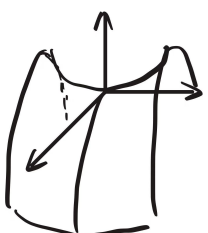
2. rank $A = 2$

$$(+, +, 0; +; 0) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + z = 0$$


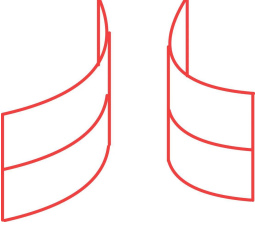
Elliptical paraboloid 椭圆抛物面

$$(+, +, 0; 0; +) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

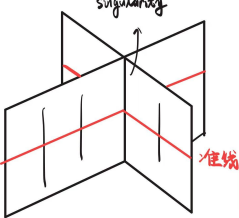

Elliptical cylinder 椭圆柱面

$$(+, -, 0; +; 0) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} + z = 0$$


Hyperbolic paraboloid/
Saddle surface 鞍面

$$\begin{cases} (+, -, 0; 0; +) & \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \\ (+, -, 0; 0; -) & \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1 \end{cases}$$


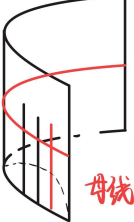
Hyperbolic cylinder

$$(+, -, 0; -; 0) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$


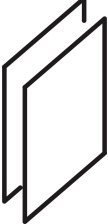
singularity

准线

3. rank $A = 1$

$$(+, 0, 0; +; 0) \quad \frac{x^2}{a^2} + y = 0$$


母线

$$(+, 0, 0; 0; +)$$


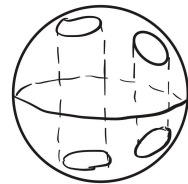
Double plane

$$(+, 0, 0; 0; 0)$$

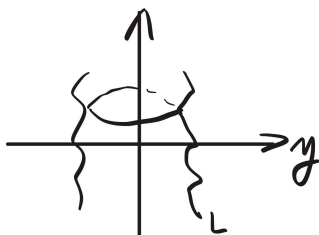
$$(+, 0, 0; 0; -) \rightarrow \emptyset$$

Intersection of Quadric Surface and Plane

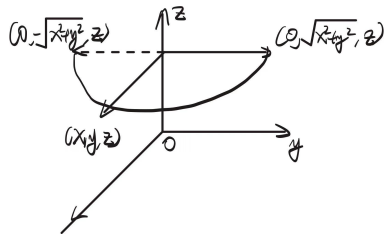
$$\begin{cases} f(\vec{x}) = \vec{x}^T A \vec{x} + \vec{b}^T \vec{x} + c = 0 \\ \vec{n}^T \vec{x} + d = 0 \end{cases} \quad \begin{cases} x^2 + y^2 + z^2 = 1 \\ (x - a)^2 + y^2 = r^2 \end{cases}$$



Surface of Revolution 旋转



$F(y, z) = 0 \rightarrow$ curve L revolved along Z axis $\Rightarrow$ Surface implicit equation?

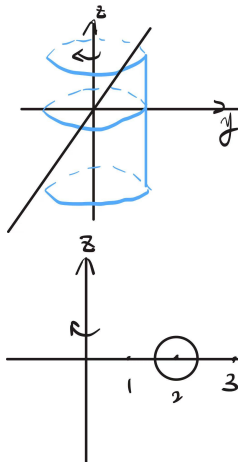


$$\Rightarrow F\left(\pm\sqrt{x^2+y^2}, z\right)=0$$

$\Leftrightarrow (x, y, z)$ lies on the surface of revolution

Eg.

$$L=\{(0, y, z) \mid y=z\}$$

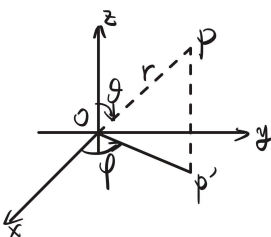


$$\begin{aligned} \pm\sqrt{x^2+y^2}=z &\rightsquigarrow x^2+y^2-z^2=0 \\ x^2+y^2+4\pm 4\sqrt{x^2+y^2}+z^2=1 &\Rightarrow (x^2+y^2+z^2+3)^2=16(x^2+y^2) \\ (y-2)^2+z^2=1 &\rightsquigarrow (\pm\sqrt{x^2+y^2}-2)^2+z^2=1 \end{aligned}$$

8.4 坐标变换和其他常用坐标系

8.4.1 坐标变换

8.4.2 Other Coordinate System

1.  $(r, \theta, \varphi) \begin{cases} r \in [0, +\infty) \\ \theta \in [0, \pi] \\ \varphi \in [0, 2\pi) \end{cases} \rightsquigarrow (x, y, z)^T = (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta)$

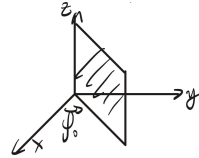
1:1 on the “open point”

$$x^2 + y^2 = r^2 \sin^2 \theta \quad z = r \cos \theta$$

$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \arctan \frac{\sqrt{x^2 + y^2}}{z} \\ \varphi = \arctan \frac{y}{x} \end{cases}$$

• $r = r_0$ 

• $\theta = \theta_0$ 

• $\varphi = \varphi_0$ 

Curvilinear coordinate system

2. Cylindrical coordinate (r, θ, z)

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases} \iff \begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \arctan \frac{y}{x} \\ z = z \end{cases}$$

9 多变量函数的微分学

9.1 多变量函数及其连续性

9.1.1 平面上的点集

9.1.2 多变量函数

9.1.3 多变量函数的极限

9.1.4 多变量函数的连续性

9.2 多变量函数的微分

9.2.1 多变量函数的偏微商

9.2.2 多变量函数的可微性

9.2.3 方向导数与梯度

9.2.4 复合函数的微分和一阶微分形式不变性

9.2.5 向量值函数的微商和微分

9.3 隐函数定理和逆映射定理

9.3.1 隐函数的存在性和微商

9.3.2 从微分的角度看隐函数定理

9.3.3 逆映射的微商

9.4 空间曲线与曲面

9.4.1 参数曲线

9.4.2 参数曲面

9.4.3 隐式曲线和隐式曲面

9.5 多变量函数的 Taylor 公式与极值

9.5.1 二元函数的微分中值定理

9.5.2 二元函数的 Taylor 公式

笔记

9.5.3 二元函数的极值

笔记

9.5.4 条件极值

笔记

9.6 向量场的微商

9.6.1 向量场

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Def (定义标题). 定义内容...

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证明. 证明过程...

□

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...

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